

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Sixth Semester of B. Tech. (CE) Examination

May 2014

CE305 Computer Network and Internetworking Layers

Date: 06.05.2014, Tuesday

Time: 10:00 a.m. To 01:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Figures to the right indicate full marks.

SECTION – I

Q - 1 Answer the following questions. [07]

- (a) What is the difference between error correction and error detection methods? Explain ARQ (Automatic Repeat Request). [03]
- (b) Differentiate OSI model and TCP/IP protocol suite. [04]

Q - 2 Answer the following questions. [14]

- (a) Explain Go back-N ARQ protocol for all cases. Compare it with Selective Repeat protocol. [07]
- (b) Explain Link state routing algorithm in detail. [07]

OR

Q - 2 Do as directed. [14]

- (a) Explain Sliding window protocol in datalink layer. [07]
- (b) Explain Distance vector routing algorithm with example. [07]

Q - 3 Do as Directed. [14]

- (a) Differentiate following: [04]
 - (i) X.25 and Frame relay.
 - (ii) Connection oriented and connection less services.
- (b) Draw and explain IP version 4 header format. [05]
- (c) What is Tunneling? Explain it with example. [05]

OR

Q - 3 Answer the following questions. [14]

- (a) Explain HDLC frame format and control fields in detail. [07]
- (b) Explain ATM reference model with appropriate diagram. [07]

SECTION – II

Q - 4 (a) Attempt following. [07]

1. What is binary exponential backoff algorithm? Explain in detail. [03]
2. List out socket primitives for TCP. [04]

Q - 5 (a) Explain the hidden terminal problem and exposed terminal problem of Wireless LAN for multiple access. Also explain the solution for that. [07]

- (b) Do as directed. [07]
 1. Draw frame format for Ethernet and explain each term in detail. [04]

2. **Differentiate:** TCP and UDP.

[03]

OR

Q - 5 (a) Explain Carrier sense multiple access protocol in detail.

[05]

(b) Draw and explain TCP header in detail.

[05]

(c) Explain UDP header in detail.

[04]

Q - 6 (a) Explain three-way handshake for establishing a connection at transport layer.

[07]

(b) Briefly explain architecture of (WWW) World Wide Web.

[07]

OR

Q - 6 (a) Explain upward and downward multiplexing with figure.

[07]

(b) Explain DNS (Domain Name System) Server in detail.

[07]
